Abstract

Emotion detection and analysis is challenging and emerging issues in natural language processing (NLP). Detecting an individual's emotional state from textual data is an active study area. Nowadays, it has become essential to understand people's emotions at a fine-grained level for system enhancement. The study of emotions can benefit many applications in various fields, including neuroscience, data mining, e-learning, information filtering systems, cognitive science, understanding expressed emotions, human-computer interaction, recommendation systems, and psychology. The rich text source available in social media, blogs, customer reviews, and news articles can help explore various insights in text mining, including emotions. Also, Bengali is the sixth most spoken language globally, and there is less research done for text-based emotion detection in Bengali because of a lack of resources. It is already challenging for Bangla texts due to the lack of labeled data. Being a multilingual society, many native Bengali speakers write their comments or posts in transliterated Bengali on social media platforms. Almost all the Bengali and transliterated Bengali texts on social media are unstructured and need to be structured and annotated. The unavailability of standard classifying models and lack of annotated corpus in the transliterated Bengali left this area of research still an exploring region. So developing a solid model for analyzing emotion from Bengali and Transliterated Bengali words from social media has become a necessity to strengthen the field of Bengali text-based sentiment analysis. This project proposes the step-by-step development of a low resource emotional corpus consisting of Transliterated Bangla (Banglish) data from social media platforms and approaches to Machine Learning (ML), Deep Neural Network (DNN), and Transformer based models to classify Ekman's six emotions from collected Banglish data and existing Bangla and English data. The BEmoD, ISEAR, IMDB, and SST2 dataset along with the newly developed Transliterated Bangla (Banglish) dataset was used to train the models. The Cohen's Kappa score of the Transliterated Bangla dataset is 89% which shows the agreement among annotators. Experimental results show that CNN + BiLSTM (FastText) outperforms all the DNN approaches with 67.54% accuracy, and Random Forest (Tf-Idf) outperforms all the ML approaches with 67.95% accuracy on Banglish data. Whereas BEmoD Bangla-BERT outperforms all the approaches with 62.19% accuracy. Bert model was trained using the SST2, IMDB, and ISEAR dataset with respective accuracy of 94.89%, 94.98%, and 72.17%.

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Chapter 1

Introduction

The Internet has become an inextricable aspect of people's lives in the twenty-first century. People now have instant access to information thanks to rapid advancements in the IT sector. With the expanding population of nations like Bangladesh, people utilizing social media platforms like Facebook, Instagram, WhatsApp, and Twitter have increased with the use of the Internet. It has bridged the gap in people's lives. Because of this, a social media analysis has grown in popularity around the world to understand people's reactions better to various social, economic, cultural, educational, and other activities on social media[1]. Data available in social media are unorganized since people from diverse races, cultures, and languages project their opinions, views, expressions, and ideas on the Internet. This data can be found in the form of text on the Internet and within large corporations such as Google, Yahoo, Microsoft, Apple, and others. They get to know the opinions of their employees spread across the continents in the form of customer feedback collected from emails and call centers or results from surveys conducted by organizations. This has sparked much interest in sentiment analysis research. Sentiment Analysis has emerged as a popular solution to gain deeper insights into people's emotions. Such as knowing how happy, sad, or angry our people are or predicting movie reviews, election results, and the stock market result, gathering feedback before releasing a new product and visualizing how customers feel about a product or service run by a company[2]. However, extracting emotions from grammatically erroneous texts, short messages, sarcasm in written documents, and contextual information could be backbreaking.

For resource-constrained languages, such as Bengali, analyzing these massive volumes of data to identify underlying sentiments or emotions is difficult. The difficulty occurs due to several constraints, including a lack of Bangla Language Processing tools, insufficient Bangla datasets, complex language structure, and restricted resources. We can find many datasets for the Bangla language, but most are very small and poorly organized. Also, Having a good dataset in Transliterated Bengali has become necessary as most of the Bangla text data online are now written in English and are unstructured. This project aimed to develop a good dataset for Transliterate Bengali and find data augmentation techniques to augment those data with Ekman's basic six emotions[3]. Also, we want to evaluate different emotion detection techniques like ML, DNN, and Transformer to find the suitable one for the Transliterate Bengali (Banglish) and Bangla datasets. English dataset was trained using Bert Model.

Chapter 2

Related Work

A substantial amount of research has been conducted in the field of NLP to understand emotions or sentiments from texts. Most of them are in high resource languages like English, Chinese and Arabic. Only a few researchers or NLP enthusiasts worked on a low-resource language like Bangla. Das et al.[4] are one of them. Their paper experimented with ML, CNN, and transfer-based models on a Bangla dataset of Ekman's six-class emotion. Among those approaches, XML-R gave an excellent f1 score of 69.73% on test data. Sharfuddin et al.[5] used RNN for sentiment classification from Bengali text. Their deep recurrent neural network with BiLSTM gave 85.67% accuracy. Rahman et al.[6] worked with a dataset of Bengali Sports news comments collected from newspapers and applied CNN, LSTM, and Multilayer Perceptron to predict the sentiment of the comments. Alam et al.[7], back in 2017, gave a state of the art accuracy (99.87%) in sentiment classifier. They applied CNN in a dataset consisting of Bangla comments. Alam et al.[8] did a great job in multi-domain text classification by applying to fine-tune multilingual transformer models on a Bangla text dataset. Their domain includes emotion detection, sentiment analysis, authorship attribution, and news categorization. Tripto et al.[9] extracted sentiments from Bangla texts using DL-based models. They experimented with three classes, five classes, and six class sentiment labels, and the dataset consists of Bangla, English, and Romanized Bangla comments from YouTube videos. Their accuracy was 65.97% and 54.24%, in three and five labels sentiment. Moreover, they claim their model is better for domain and language-specific texts. Along with DL approaches, a few ML approaches have been made. Tuhin et al.[10] applied two machine learning techniques, Naïve Bayes and Topical, to extract six individual emotion classes from Bangla text. Their topical approach showed better results. Rahman et al.[11], in his paper, applied five popular ML techniques (Naive Bayes, k-NN, SVM, Decision Tree, and K-Means Clustering) on a dataset consisting of Facebook comments and labeled them into six basic emotions (sadness, anger, happiness, surprise, disgust, fear). Among them, SVM showed the best accuracy of 52.98%. Another multi-class emotion prediction from Bangla text was made by Azmin et al.[12] using a Multinomial NB classifier. Their model was able to classify the text into three emotion classes (happy, sad, and angry) with 78.6% accuracy. Rush et al.[13] developed a corpus consisting of 1200 Bangla words and trained them on an SVM classifier. They were able to recognize six basic emotions (happy, sad, disgust, anger, surprise, and fear) with 73% accuracy

Chapter 3

Project Design

We have started our project by collecting pre-existing datasets and by developing our own dataset for Transliterate Bengali. A Transliterate Bengali (Banglish) dataset was developed with Ekman's six emotions due to data unavailability. Developing a dataset in Transliterated Bengali (Banglish) is a significant undertaking. The most noticeable issues are the shortage of adequate emotional text expression and unstructured text over social media. Almost 3500 data were collected via web scraping from social media platforms (Facebook, Twitter, and Youtube) and were vigorously cleaned for two months. The total data was 3044 which became 7611 after data augmentation. For Bengali few available datasets like BEmoD[14] and for English popular datasets like ISEAR[15] were used.

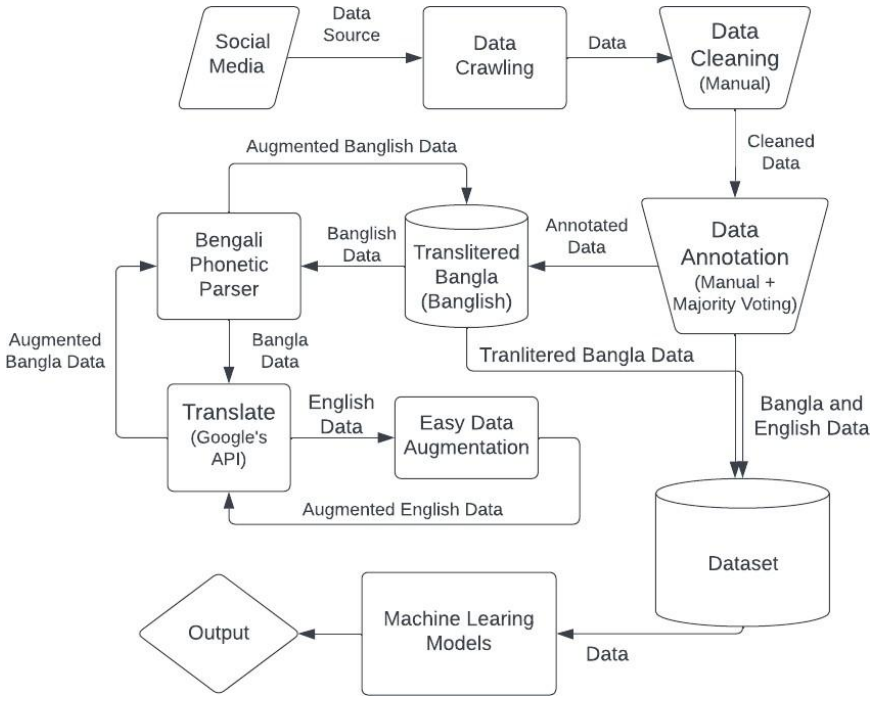


Fig 3.1: System Diagram (Detailed View)

After collecting the data we applied different embedding matrices as feature extractors and extract features from our data. Then those features were fed into the machine learning, deep learning, and transformer-based learning model. After training the models were evaluated and showed our desired output/emotion of a particular text.

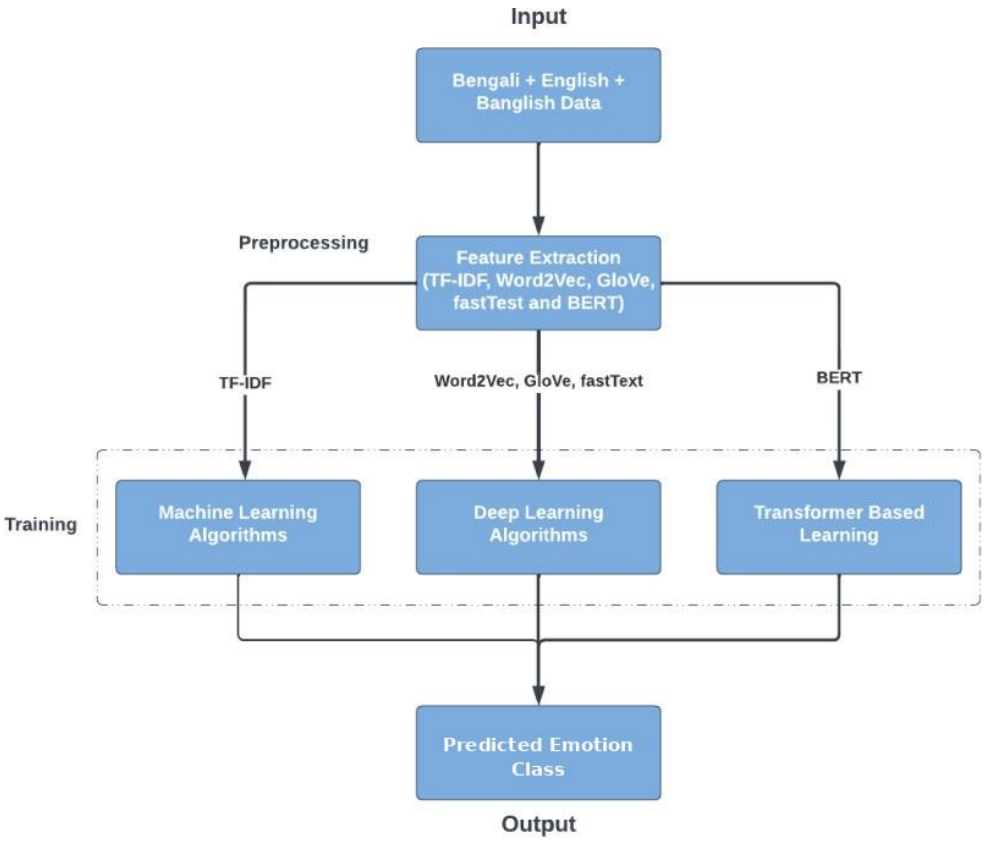


Fig 3.2: System Diagram

Chapter 4

Approach

4.1 Feature Extraction

Word2Vec, GloVe, and FastText were implemented for ML and DNN tasks. Skip-Gram model architecture was used to represent words in a multidimensional vector space. It aims to predict the surrounding context words given the center word[16][17].

TF-IDF: The term frequency-inverse document frequency (TF-IDF) is a statistical tool for measuring the importance of a word in a collection of documents[18]. This is accomplished by combining two metrics: the number of times a word appears in a document and the word's inverse document frequency over a set of documents.

Word2Vec: Word2Vec generates word vectors, which are distributed numerical representations of word features including context [19]. The vectors created by word embeddings eventually assist form the relationship between a word and another word with a similar meaning.

GloVe: The outputs of this vector representation are linear substructures of the word vector space[20], which are based on collected global word-word co-occurrence data from a corpus. GloVe was trained using a 300-dimensional embedding dimension.

FastText: The word2vec embedding mode now includes this word embedding approach. Each word is formed as an n-gram of characters[21]. FastText was learned on Skip-Gram with 300 embedded dimensions.

For all of them Word2Vec, GloVe, and FastText, pre-trained vectors are available for the Bengali language trained on generalized Bengali wiki dump data[22]. This was used to train the models.

4.2 Machine Learning Models

Initially, ML models were used to classify emotion on the Transliterated Bangla (Banglish) dataset. XGBoost, Decision Tree, SVM, Naive Bayes, Simple Gradient Descent, Random Forest, and Logistic Regression models were used using the TF-IDF text vectorizer. All the ML models were run on default setup with no changes in hyperparameter.

4.3 Deep Neural Networks

Few variations of DNN models were used to evaluate the performance of our dataset and the Bangla dataset(BEmoD). We integrated the Attention mechanism to enhance the performance of the task.

CNN-BiLSTM: Convolutional Neural Network (CNN) is tuned over the emotion corpus[23]. The trained weights from the GloVe/FastText embeddings are fed to the embedding layer to generate a sequence matrix. An Embedding layer comparable to CNN is used in the developed BiLSTM (Bidirectional Long-Short Term Memory) network. On top of two BiLSTM layers with 64 and 32 units, an embedding layer, a 1D convolutional layer with 64 filters of size three, and a 1D max-pool

layer are used, with the outputs of the BiLSTM layer sent to an output layer with 'softmax' activation. The hyperparameters that exhibited the best accuracy in CNN+BiLSTM are included in the table.

Attention: The attention method is used to improve DNNs by assigning different weights to the key elements of the context [24]. GloVe and FastText were employed as embedding layer weights in the proposed model. The attention mechanism is then employed to enable the model to pay more or less attention to certain words and sentences[25].

Table 4.1: Hyperparameters for deep neural network methods

Hyperparameters	**CNN**	**CNN+BiLSTM**	**CNN + Attention + BiLSTM/LSTM + CNN**
Base Learning Rate	-	-	1e-3
Max Learning Rate	-	-	6e-3
Learning rate mode	-	-	exp_range
Pooling type	max	max	max
Embedding dimension	300	300	300
Batch size	64	64	64
Activation function	swish, softmax	swish, softmax	Relu, softmax
Learning rate	0.001	0.001	0.001
Optimizer	adam	adam	adam
Epoch	35	35	20
Max Len	50	215	50

4.4 Transformer Based Learning

Transformers are semi-supervised machine learning models used primarily for textual data and have replaced recurrent neural networks in natural language processing. Transformer models like BERT, which Google AI Language develops, are extensively used to achieve state-of-the-art accuracy in natural language processing tasks[26]. Some pre-train BERT models like Bert Base, Multilingual Bert, and Bangla Bert models were used to classify the emotions from the text. The Huggingface transformers library was used to invoke these pre-trained models and, later, fine-tuned based on the datasets[27]. A lightweight wrapper for Keras, Ktrain was used to train these models to reduce complexity in code[28]. While training, 80% of data was used as training data, and 20% of data was used as test data.

BERT Base: The BERT Base model was trained using BookCorpus, a dataset of 11,038 unpublished books, as well as English Wikipedia [26]. On the ISEAR dataset, the 'bert-base-uncased' model was utilized. Seven emotions are represented in the ISEAR dataset. All of the datasets were trained with a maximum length of 500.

Bangla BERT: Bangla Bert is a Bengali language model that employs BERT's mask language modeling. This model used the bert-base-uncased model architecture [29] and utilized the BNLP package to train the Bengali sentence piece model with vocab size 102025. On the BEmoD dataset, the 'sagorsarker/Bangla-bert-base' mode was employed. The dataset was trained using a batch size of 6 and a maximum length of 500.

Multilingual BERT: Multilingual Bert, or m-BERT, is a transformer model with about 110 million parameters[26] and has been pre-trained in 104 languages. The m-Bert model was trained using an automated approach to construct inputs and labels from the raw texts, with no humans recognizing them in any way. The BEmoD dataset was fine-tuned using the bert-base-multilingual-cased command. The dataset was trained using a batch size of 6 and a maximum length of 500.

Table 4.2: Hyperparameters for transformer-based learning methods

Hyperparameters	**BERT Base**	**Bangla BERT**	**Multilingual BERT**
Learning rate mode	one_cycle	one_cycle	one_cycle
Max length	500	500	500
Batch size	6	6	6
Learning rate	2e-5	2e-5	2e-5
Epoch	5	5	5

Chapter 5

Experimental Analysis

5.1 Challenges Faced

5.1.1 Data Cleaning

After collecting the data, it was cleaned manually. In manual cleaning spellings and incomplete words were corrected, and special characters and irrelevant punctuation marks were removed. We used NLTK tool kit to remove the noisy words and foreign languages. After cleaning, the total data was 3044. The example of before and after cleaning is given below.

5.1.2 Data Annotation and Quality

Four undergraduate students working in NLP were assigned to annotate the data individually. After annotation, majority voting techniques were used where reviewing four students' four annotations on one single text, one of the annotations was finalized. For example, ' amar oporadh ami hindu ' for this sentence, three of the students annotated this as 'angry ' and one as 'fear'. So using majority voting, we finalized this text as an angry annotation. Also, some texts couldn't use majority voting because of the equal ratio. For those texts, a discussion was held. If still, it was not very clear, then it was discarded. Cohen's Kappa[30] score of the dataset was 89%.

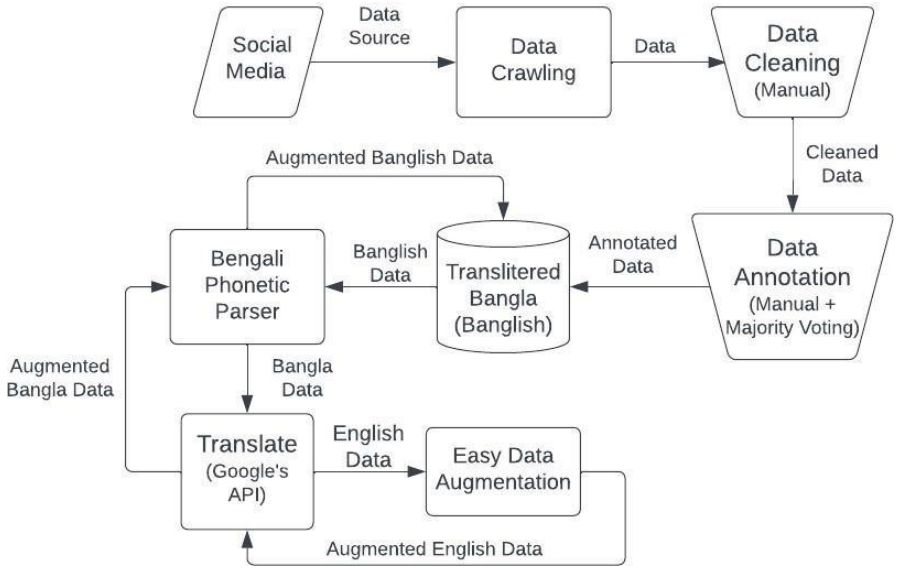


Fig 5.1: Dataset development model

5.1.3 Data Augmentation

Easy Data Augmentation (EDA) technique was applied on 3004 labeled Transliterate Bengali (Banglish) datasets [31]. There weren't any augmentation techniques for Banglish directly. First, to do augmentation, we converted the Banglish dataset to Bengali using the Bangla Phonetic Parser[32]. Then Google's API was used to convert the Bangla dataset to English. Then EDA was applied to this English dataset for boosting performance on text classification tasks. Easy Data Augmentation consists of four simple but powerful operations with CNN and RNN. These are Synonym Replacement, Random Insertion, Random Swap, and Random Deletion.

All four techniques were used to augment the Banglish dataset. After augmentation, the total data was 9008 English data which was labeled accordingly. Again we convert this English dataset to Bangla and Bangla to Transliterate Bangla (Banglish). But this time, while converting Bangla to Banglish, the Bangla parse technique didn't work perfectly. It made the sentence, but most of the words were not understandable. Moreover, some of this data is so wrecked that we need to remove the whole sentence. After manually cleaning, the total Transliterate Bangla data was 7611.

5.1.4 Data Statistics

Initially 3044 data was collected from Facebook, Youtube, and Twitter after cleaning. Where 'surprise' is 537, 'happy' is 530, 'sad' is 515, 'fear' is 494, 'angry' is 491, and 'disgust' is:477. After doing EDA and again manual cleaning the dataset became larger and the total data became 7611 with 'surprise' becoming 1319, 'happy' 1305, 'sad' 1288, 'angry' 1272, 'fear' 1231, and 'disgust' 1196

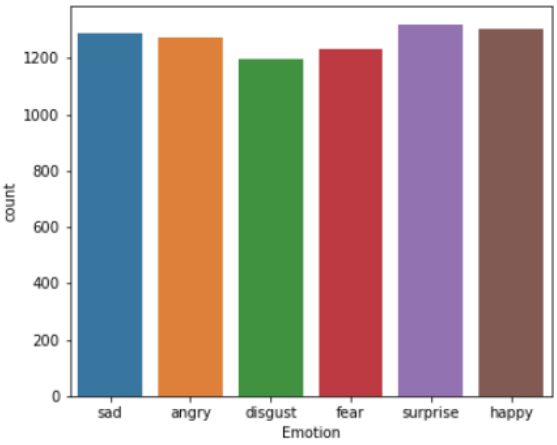


Fig 5.2: Banglish dataset after Augmentation

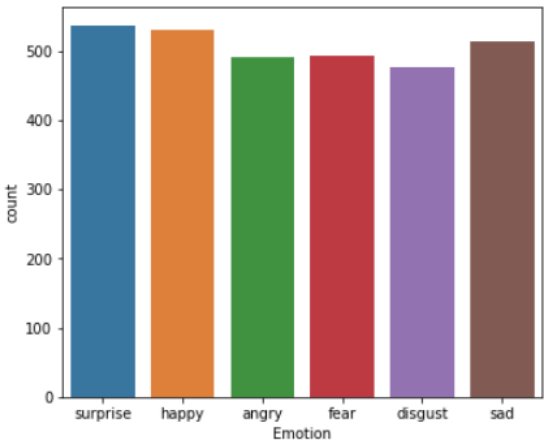


Fig 5.3: Banglish dataset Before Augmentation

ISEAR: The ISEAR study gathered data from a broad group of psychologists from all over the world over a long period in the 1990s. This dataset contained reports on seven emotions based in 37 countries on all five continents[15].ISEAR is a balanced dataset with seven emotions.

Emotion	Number of Data
joy	1081
fear	1081
anger	1071
sadness	1067
disgust	1067
shame	1054
guilt	1052

Fig 5.4: ISEAR dataset description

BEmoD: BEmoD emotional dataset and development method for analyzing emotion in Bengali texts, including data, crawling, pre-processing, labeling, and verification. BEmoD contains 5200 texts labeled into six basic emotional categories: anger, fear, surprise, sadness, joy, and disgust, respectively[14]. The data was collected from the Internet using crawling from various sites. There are 5619 Bangla data in the BEmoD dataset, which are divided into six emotions.

Emotion	Number of Data
disgust	1398
sadness	1061
joy	1022
fear	783
anger	692
surprise	663

Fig 5.5: BEmoD dataset description

5.2 Result Analysis

This section reviews all the models (ML and DNN models) tried on our Transliterated Bengali(Banglish), Bangeli, and English dataset to classify emotions from the text. For Banglish, the models were applied in two parts before augmentation and after augmentation to observe the difference between two different sized datasets. For now, we chose accuracy to determine a model's performance. However, other matrices (precision, recall,f1) were also considered to evaluate the performance. The Table shows the matrices of all the models. On a larger dataset (after annotation), RF (ML model) outperformed all the models with 67.95% accuracy. Next in line was the DNN model CNN+ BiLSTM(FastText) with 67.54% accuracy. ML models performed quite well. SVM, Simple Gradient Descent, and LR were better than others with 67.30%, 67.03%, and 67.49% accuracy. All DNN models performed quite well except CNN + Attention + LSTM + CNN (GloVe). Its accuracy was significantly lower than others. However, none of the DNN models could beat RF's accuracy. Nevertheless, with a smaller dataset(before annotation), CNN-BiLSTM with GloVe (DNN model) outperformed other methods concerning all the evaluation matrices. It achieved 56.77% accuracy. CNN-BiLSTM(FastText) was close in terms of performance with 55.08% accuracy. However, ML could not meet our expectations. Among ML models, LR performed well with 53.20% accuracy. The second best was RF with 52.05% accuracy.

	precision	recall	f1-score	support
anger	0.62	0.63	0.63	268
fear	0.66	0.71	0.69	224
joy	0.83	0.78	0.80	275
disgust	0.66	0.73	0.69	262
sadness	0.67	0.42	0.52	264
surprise	0.62	0.78	0.69	229
accuracy			0.67	1522
macro avg	0.68	0.68	0.67	1522
weighted avg	0.68	0.67	0.67	1522

Fig 5.6: Precision, recall, f1-score matrix of CNN+BiLSTM(Word2Vec) on Banglish

	precision	recall	f1-score	support
joy	0.87	0.89	0.88	204
sadness	0.70	0.77	0.73	209
fear	0.76	0.78	0.77	221
anger	0.69	0.64	0.66	219
disgust	0.72	0.73	0.72	222
shame	0.64	0.61	0.63	215
guilt	0.66	0.65	0.66	205
accuracy			0.72	1495
macro avg	0.72	0.72	0.72	1495
weighted avg	0.72	0.72	0.72	1495

Fig 5.7: Precision, recall, f1-score matrix of BERT Base on ISEAR

	precision	recall	f1-score	support
joy	0.65	0.67	0.66	199
sadness	0.61	0.59	0.60	229
surprise	0.54	0.56	0.55	135
disgust	0.65	0.69	0.67	275
anger	0.48	0.45	0.47	134
fear	0.75	0.70	0.73	152
accuracy			0.62	1124
macro avg	0.61	0.61	0.61	1124
weighted avg	0.62	0.62	0.62	1124

Fig 5.8: Precision, recall, f1-score matrix of Bangla BERT on BEmoD

It was monitored throughout the experiments that after doing the annotation the accuracy of the ML and DNN models improved on an average of 15%-20%. This is because in the overall annotation process EDA increased the size of the data and manual cleaning improved the quality. Now, DNN along with transformer-based models were applied to classify the BEmoD dataset. In DNN, CNN + Attention + BiLSTM + CNN (GloVe) and CNN + Attention + LSTM + CNN (FastText) outperformed other methods with 60.28% accuracy. However, after employing transformer-based models, it observed a slight increase in scores. Among transformer-based models, Bangla-BERT achieved an accuracy of 62.19% which is the highest among all models.

However, BERT models need much computational power, so a GPU is needed to run this model. BERT models also take much time to train. In comparison, CNN BiLSTM models do not need much computational power and are faster than BERT. Also, BERT does not perform as well as English for low-resource languages like Bangla. Using BERT Base, the English language got an accuracy of 95%, and Bangla got an accuracy of 62.19%. The reason for the lower accuracy in the Bangla dataset is that the Bangla language does not have a large corpus.

Table 5.1: Machine learning models with accuracies

Model	**Accuracy(%) Before Augmentation**	**Accuracy(%) after Augmentation**
XGBoost (Tf-IDF)	45.15	54.95
Decision Tree (Tf-IDF)	49.58	65.39
SVM (Tf-IDF)	51.88	67.30
Naive Bayes (Tf-IDF)	50.57	59.15
Simple Gradient Descent (Tf-IDF)	51.06	67.03
Random Forest (Tf-IDF)	52.05	67.95
Logistic Regression (Tf-IDF)	53.20	67.49

Table 5.2: BERT Accuracy on English Dataset

Dataset Name	**Learning Rate**	**Epoch**	**Accuracy**
ISEAR	2e-5	5	71.17%

Table 5.3: Deep neural network models with BEmoD dataset

DNN Models	Accuracies (%)
CNN + Attention (Glove)	57.28
CNN + Attention + BiLSTM + CNN (GloVe)	60.28
CNN + Attention + LSTM + CNN (GloVe)	56.60
CNN + Attention + BiLSTM + CNN (FastText)	58.76
CNN + Attention + LSTM + CNN (FastText)	60.28
CNN + BiLSTM (GloVe)	56.64
CNN + BiLSTM (FastText)	56.32
Bangla Bert	62.19
m-Bert	56.00

Table 5.4: DNN model accuracies before and after augmentation

Models	Accuracies(%) Before Augmentation	Accuracies(%) After Augmentation
CNN (Word2Vec)	48.77	67.15
CNN + BiLSTM (GloVe)	56.72	67.41
CNN + BiLSTM (FastText)	55.08	67.54
CNN + BiLSTM (Word2Vec)	35.46	67.73
CNN + Attention + BiLSTM + CNN (Glove)	48.76	66.36
CNN + Attention + BiLSTM + CNN (FastText)	51.88	66.90
CNN + Attention + LSTM + CNN (Glove)	41.37	66.64
CNN + Attention + LSTM + CNN (FastText)	47.61	66.51

5.3 Error Analysis

Among DNN models CNN-BiLSTM(Word2vec) showed comparatively better performance. That's why we did a thorough error analysis with a confusion matrix of that particular model.

	precision	recall	f1-score	support
anger	0.62	0.63	0.63	268
fear	0.66	0.71	0.69	224
joy	0.83	0.78	0.80	275
disgust	0.66	0.73	0.69	262
sadness	0.67	0.42	0.52	264
surprise	0.62	0.78	0.69	229
accuracy			0.67	1522
macro avg	0.68	0.68	0.67	1522
weighted avg	0.68	0.67	0.67	1522

Fig 5.9: Precision, recall, f1-score matrix of CNN+BiLSTM(Word2Vec) on Banglish

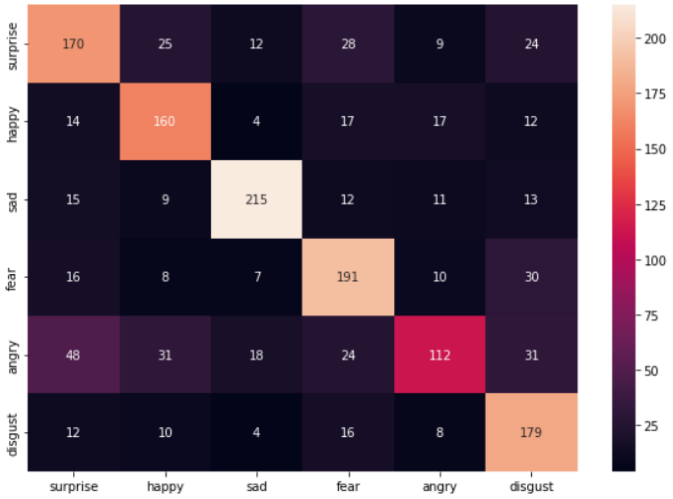


Fig 5.10: Confusion matrix of CNN+BiLSTM(Word2Vec) model on Banglish

The graph depicts a class-by-class proportion of estimated labels. The matrix reveals that a small number of data points were incorrectly categorized. For example, 25 samples among 1090 of the 'surprise' class were predicted as 'happy'. In the 'sad' class, 12 data out of 1024 were misclassified as 'fear'. In the 'anger' class, the misclassification ratio is the highest. That means, 48 data out of 1004 in the 'anger' class were incorrectly classified as 'surprise'. The error analysis disclose that the 'sad' class had the highest rate of correct classification while 'angry' achieved the lowest of all.

```
array([[134,  20,  20,  14,   9,   2],
       [ 24, 134,  11,  24,  21,  15],
       [ 20,  15,  75,  11,   2,  12],
       [ 18,  25,  15, 189,  23,   5],
       [  8,   9,   4,  51,  60,   2],
       [  2,  17,  15,   2,   9, 107]], dtype=int64)
```

Fig 5.11: Confusion Matrix of Bangla BERT on BEmoD

```
array([[181,   7,   4,   2,   0,   7,   3],
       [  6, 160,   9,  11,   7,   7,   9],
       [  7,  17, 172,   5,   7,   6,   7],
       [  2,  17,   7, 140,  30,  13,  10],
       [  2,  10,  13,  23, 161,  12,   1],
       [  7,   6,  12,  10,  11, 131,  38],
       [  2,  13,   8,  12,   8,  28, 134]], dtype=int64)
```

Fig 5.12: Confusion Matrix of BERT Base on ISEAR

Whereas in Bangla Bert on BEmoD we found out the ‘disgust’ class achieved the highest accuracy rate and ‘anger’ class gained the lowest accuracy rate. And for BERT Base on ISEAR ‘joy’ class achieved the highest accuracy rate and the ‘shame’ class gained the lowest.

There are many possible reasons for inaccurate predictions e.g class imbalance in the dataset. So by creating a balanced dataset prediction error could be reduced to minimal. Also, a few words could be used to mean multiple things, for example, hate words can express both 'disgust' as well as 'anger'.

Chapter 6

Conclusion

This research is encouraged on detecting emotion from Transliterated Bengali (Banglish). Developing a good dataset due to the unavailability of data and doing data augmentation techniques for low resource language like Bengali was the main contribution that led the way towards emotion detection from Transliterated Bengali(Banglish). Various ML and DNN techniques were used to investigate the performance of the data in classifying six classes of emotion from Banglish words. The Cohen's Kappa score of 0.89 reflects the quality of the dataset. Furthermore, among ML models, Random Forest with TF-IDF got the highest accuracy of 67.95%, and on the DNN part, CNN+ BiLSTM with Word2Vec got the best accuracy of 67.54%. The dataset quality and size improved after augmentation and manual cleaning, which led to a 15%-20% improvement in accuracy on both ML and DNN models. Currently, in NLP, state-of-the-art accuracy is achieved through transformed-based models like Bert[25]. So having a pre-trained model for Banglish data for Bert will help Banglish data perform even better. DNN models and Transformer Learning were also applied to the BEmoD dataset. CNN + Attention + BiLSTM + CNN (GloVe) and CNN + Attention + LSTM + CNN (FastText) outscored other techniques in all assessment parameters in DNN with an accuracy of 60.28%. In Transformer Learning, Bangla Bert outperformed the DNN architectures with an accuracy of 62.19%, but to get that 2% accuracy, Transfer Learning takes more time and computational power than DNN based models. BERT also worked significantly better on large English language datasets like IMDB and SST2 datasets, with an accuracy of almost 95%, whereas our Bangla language dataset gave 62.19% accuracy. That is why we prefer DNN models over Transformer models like BERT for our resource-constrained datasets.

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